Department Of Mathematics

SVNIT,Surat

Linear Algebra and Statistics (MA- 106)

Tutorial-8-Joint, marginal, Conditional -PMF and PDF, Covariance, Correlation, Linear Regression)

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1) Consider two random variables X and Y with joint PMF given in Table 5.3.

a. Find $P(X \leq 2, Y \leq 4)$.
b. Find the marginal PMFs of X and Y.
c. Find $P(Y = 2|X = 1)$.
d. Are X and Y independent?

	$Y = 2$	$Y = 4$	$Y = 5$
$X = 1$	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{24}$
$X = 2$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{8}$
$X = 3$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$

2) The joint probability function of two discrete random variables X and Y is given by $f(x,y) = cxy$ for x=1,2,3 and y= 1,2,3 and equals zero otherwise. Find:

a. The constant c.
b. $P(X = 2, Y = 3)$.
c. $P(1 \leq X \leq 2, Y \leq 2)$.
d. $P(X \geq 2)$.
e. $P(Y < 2)$.
f. $P(X = 1)$.
g. $P(Y = 3)$.

3) For the random variables of Problem 2, find the marginal probability function of X and Y. Determine whether X and Y are independent.

4)

Let X and Y be continuous random variables having joint density function

$$f(x,y) = \begin{cases} c(x^2+y^2) & 0 \le x \le 1, \quad 0 \le y \le 1 \\ 0 & otherwise \end{cases}$$

Determine:

a. The constant c.

b. $P\left(X < \frac{1}{2}, Y > \frac{1}{2}\right)$

c. $P\left(\frac{1}{4} < X < \frac{3}{4}\right)$.

d. $P\left(Y < \frac{1}{2}\right)$.

e. Whether X and Y are independent.

5) The joint probability mass function of X and Y is given below.

$y \backslash x$	-1	$+1$
0	$\frac{1}{8}$	$\frac{3}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$

Find the correlation coefficient of (X, Y).

6) Two random variables X and Y have the joint density.

$$f(x, y) = \begin{cases} 2-x-y, & 0 < x < 1,\ 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$ Show that $\text{cov}(X, Y) = \frac{-1}{144}$.

7) Suppose that the 2D RVs (X, Y) has the joint p.d.f.

$$f(x, y) = \begin{cases} x+y, & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$ Obtain the correlation co-efficient between X and Y.

8) If (X, Y) is a two-dimensional random variable uniformly distributed over the triangular region R bounded by y = 0, x = 3, and y = 4/3 x. Find the correlation coefficient r_{xy}.